

HKCEE 1987 Mathematics II

Label 1 $\left(\frac{x+1}{x}\right)^2 - \left(\frac{x-1}{x}\right)^2 =$

- A. 2
- B. $\frac{x}{4}$
- C. $\frac{2}{x^2}$
- D. $\frac{4}{x^2}$
- E. 0

Label 2 If $\frac{1}{x} - \frac{1}{y} = \frac{1}{z}$, and $x = \frac{1}{2}$, $z = \frac{1}{3}$, then $y =$

- A. $-\frac{1}{6}$
- B. $\frac{1}{6}$
- C. 5
- D. 6
- E. $\frac{1}{6}$

Label 3 If $a = \frac{b+3cd}{b-3cd}$, then $c =$

- A. $\frac{a}{6d}$
- B. $\frac{b}{3d}$
- C. $\frac{b(a-1)}{6d}$
- D. $\frac{b(a+1)}{a-1}$
- E. $\frac{b(a-1)}{3d(a+1)}$

Label 4 The radii of two solid spheres made of the same material are in the ratio 2 : 3. If the smaller sphere weight 16 kg, then the larger one weighs

- A. 24 kg
- B. 36 kg
- C. 48 kg
- D. 54 kg
- E. 60 kg

Label 5 Given that $x \neq 0$ and $-x, x, 3x^3$ are in G.P., find x .

- A. $-\frac{1}{3}$
- B. $\frac{1}{3}$
- C. $\sqrt{3}$
- D. $\frac{1}{3}$
- E. 1

Label 6 If $x + \frac{1}{x} = 1 + \sqrt{2}$, then $x^2 + \frac{1}{x^2} =$

- A. 1
- B. 3
- C. $1 + 2\sqrt{2}$
- D. $2 + 2\sqrt{2}$
- E. $3 + 2\sqrt{3}$

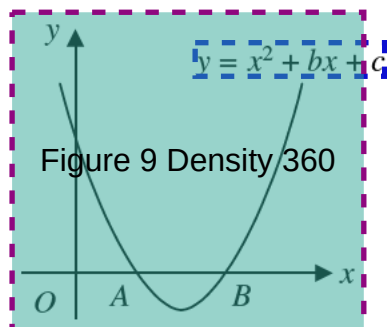
Label 7 If $3^{2k} + 1 = 3^{2k} + 6$, then $k =$

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 1
- D. $\frac{1}{2}$
- E. 3

Label 8 When the expression $x^2 + px + q$ is divided by $x + 1$, the remainder is 4. Find the value of $2p - 2q + 1$.

- A. -3
- B. -5
- C. -7
- D. -9
- E. It cannot be determined.

Label 9



In the figure, the graph of $y = x^2 + bx + c$ cuts the x-axis at A and B. $OA + OB =$

- A. b
- B. c
- C. -b
- D. -c
- E. $-\frac{b}{c}$

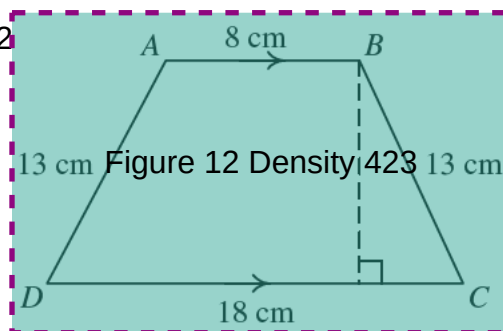
Label 10 If $f(x) = x^2 + 1$, then $f(x - 1) =$

- A. x^2
- B. $x^2 - 1$
- C. $x^2 + 2$
- D. $x^2 - 2x$
- E. $x^2 - 2x + 2$

Label 11 If $\log_{10}x, \log_{10}y, \log_{10}z$ are in A.P., then

- A. $y = 10^{\frac{x+z}{2}}$
- B. $y = \frac{x+z}{2}$
- C. $y^2 = x + z$
- D. $y^2 = xz$

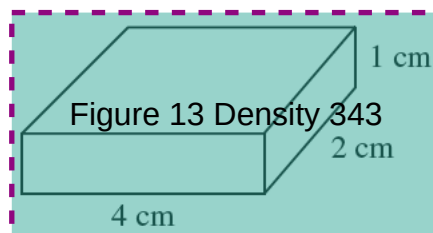
Label 12 E. $y = 10^{\sqrt{x}}$



ABCD is a trapezium in which $AB \parallel DC$, $AB = 8$ cm, $DC = 18$ cm, $AD = BC = 13$ cm. Find the area of the trapezium.

- A. 156 cm^2
- B. 169 cm^2
- C. 216 cm^2
- D. 312 cm^2
- E. 338 cm^2

Label 13



A solid rectangular iron block, $4 \text{ cm} \times 2 \text{ cm} \times 1 \text{ cm}$, is melted and recast into a cube. The decrease in the total surface area is

- A. 1 cm^2
- B. 2 cm^2
- C. 3 cm^2
- D. 4 cm^2
- E. 5 cm^2

Label 14

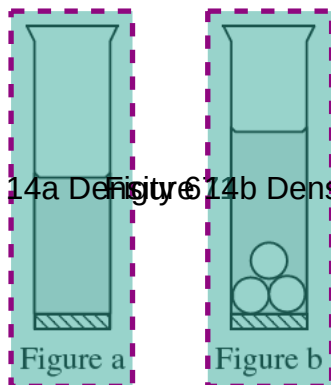


Figure 14a Density 614b Density 656

Figure a shows a circular measuring cylinder 4 cm in diameter containing water. Three iron balls, each of diameter 2 cm, are dropped into the cylinder as shown in Figure b. What is the rise in the water level?

- A. $\frac{1}{4}$ cm
- B. $\frac{1}{3}$ cm
- C. $\frac{1}{2}$ cm
- D. 1 cm
- E. 2 cm

Label 15 Find, correct to the nearest dollar, the compound interest on \$10 000 at 8% p.a. for 4 years, compounded half-yearly.

- A. \$3200
- B. \$3605
- C. \$3686
- D. \$13 200
- E. \$13 686

Label 16 If the selling price of 5 pens is the same as the cost price of 6 pens, the percentage profit in selling a pen will be

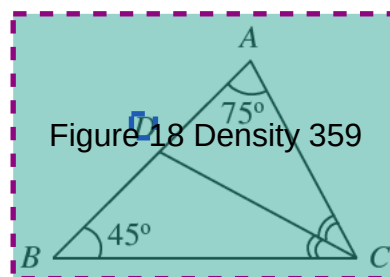
- A. $16\frac{2}{3}\%$
- B. 20%
- C. 60%

- D. $116\frac{2}{3}\%$
- E. 120%

Label 17 The circumference of a circle is 6π cm. The length of an arc of the circle which subtends an angle of $\frac{1}{3}$ radian at the centre is

- A. 1 cm
- B. $\frac{3}{2}$ cm
- C. 2 cm
- D. π cm
- E. 2π cm

Label 18



In the figure, $\angle A = 75^\circ$, $\angle B = 45^\circ$ and CD bisects $\angle ACB$. $\frac{BD}{CD}$

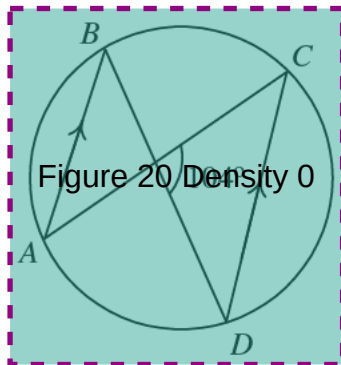
- A. 2
- B. $\frac{1}{\sqrt{2}}$
- C. $\sqrt{2}$
- D. $\sqrt{\frac{2}{3}}$
- E. $\sqrt{\frac{3}{2}}$

Label 19 A rectangle is 6 cm long and 8 cm wide. The acute angle between its diagonals, correct to the nearest degree is

- A. 37°
- B. 41°

- C. 49°
D. 74°
E. 83°

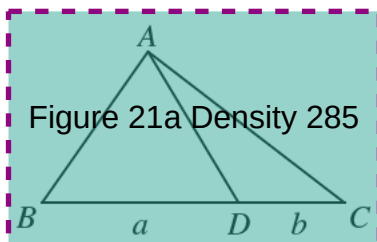
Label 20



In the figure, chords AC and BD meet at E and $AB \parallel DC$. If $\angle CED = 104^\circ$, find $\angle ABD$.

- A. 76°
B. 52°
C. 38°
D. 14°
E. It cannot be determined.

Label 21



In the figure, $BD = a$, $DC = b$ and the area of $\triangle ABC$ is s .

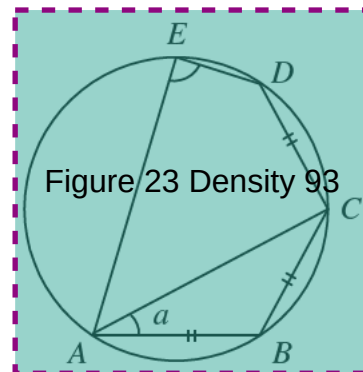
- A. $\frac{s(a+b)}{a}$
B. $\frac{s(a+b)}{b}$
C. $\frac{s(a+b)^2}{a^2}$
D. $\frac{s(a+b)^2}{b^2}$
E. $\frac{s(a^2+b^2)}{a^2}$

Figure 21b Density 497

Label 22 The real number π is

- A. $\frac{22}{7}$
B. 3.1416.
C. the ratio of the area of a circle to the square of its diameter.
D. the ratio of the circumference of a circle to its radius.
E. the ratio of the circumference of a circle to its diameter.

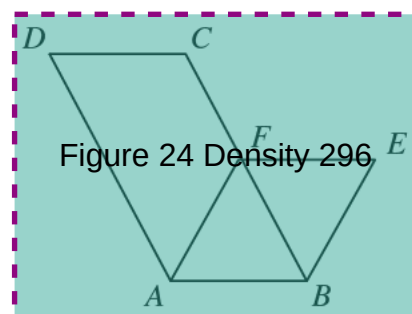
Label 23



In the figure, AB , BC and CD are three equal chords of a circle. If $\angle BAC = a$, then $\angle AED =$

- A. $2a$.
B. $3a$.
C. $90^\circ - a$.
D. $180^\circ - 2a$.
E. $180^\circ - 3a$.

Label 24



In the figure, $ABCD$ and $ABEF$ are parallelograms. $\frac{\text{Area of } ABCD}{\text{Area of } ABEF} =$

- A. $\frac{AD}{AF}$
 B. $\frac{BC}{BF}$
 C. $\frac{BC}{EE}$
 D. $\frac{AD^2}{AE^2}$
 E. $\frac{BC^2}{EF^2}$

Label 25
25

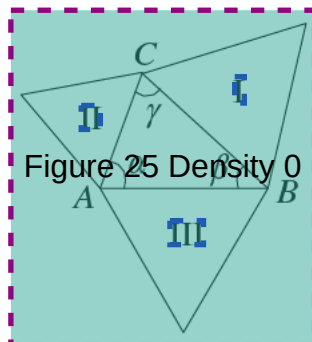


Figure 25 Density 0

In the figure, I, II and III are equilateral triangles.
 Area of I : Area of II : Area of III =

- A. $\alpha : \beta : \gamma$.
 B. $\sin \alpha : \sin \beta : \sin \gamma$.
 C. $\sin^2 \alpha : \sin^2 \beta : \sin^2 \gamma$.
 D. $\cos \alpha : \cos \beta : \cos \gamma$.
 E. $\cos^2 \alpha : \cos^2 \beta : \cos^2 \gamma$.

Label 26
26 Which of the following straight lines divide(s) the circle $(x - 1)^2 + (y + 1)^2 = 1$ into two equal parts?

- I. $x - y - 2 = 0$
 II. $x + y + 2 = 0$
 III. $x - y + 2 = 0$

- A. I only
 B. II only
 C. III only
 D. I and II only
 E. II and III only

Label 27
27 The equation of a circle is $x^2 + y^2 - 4x + 2y + 1 = 0$. Which of the following is/are true?

- I. The centre is $(-2, 1)$.
 II. The radius is 2 units.
 III. The circle intersects the y-axis at two distinct points.

- A. I only
 B. II only
 C. III only
 D. I and II only
 E. II and III only

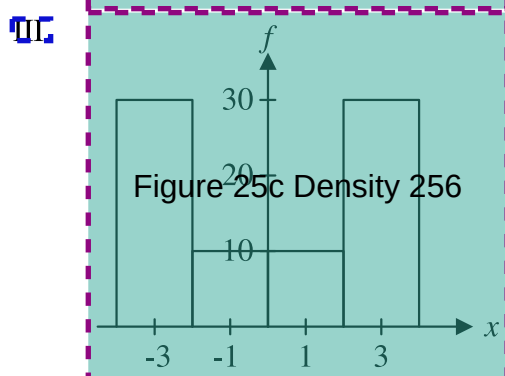
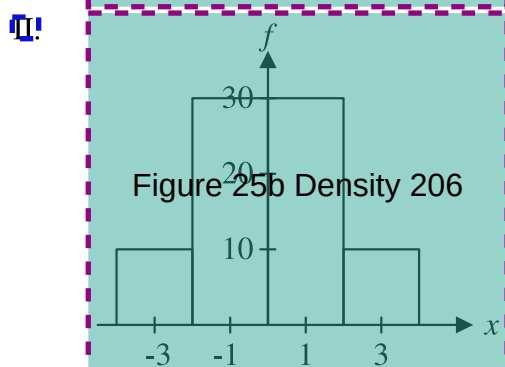
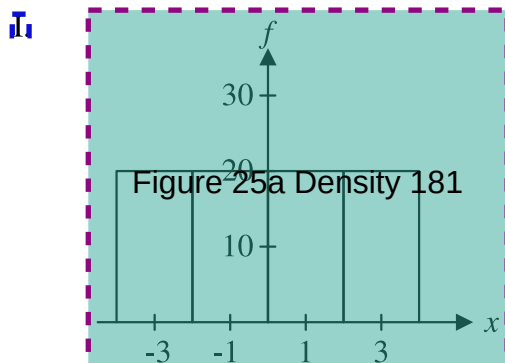
Label 28
28 Two perpendicular lines $kx + y - 4 = 0$ and $x - 2y + 3 = 0$ intersect at the point (h, k) . Find h and k .

- A. $h = -7, k = -2$
 B. $h = -2, k = \frac{1}{2}$
 C. $h = 1, k = 2$
 D. $h = -4, k = -\frac{1}{2}$
 E. $h = -3, k = 2$

Label 29
29 If the median of the 5 different integers 2, 7, 10, x , $2x - 3$ is 7, then $x =$

- A. 3
 B. 4
 C. 5
 D. 6
 E. 8

Label 30
30 The figures show the histograms of the three frequency distributions. Arrange their standard deviations in ascending order of magnitude.



- A. I, II, III
 B. I, III, II
 C. II, I, III
 D. II, III, I
 E. III, II, I

Label 31
 31. One letter is taken from each of the words “MAN” and “ART” at random. Find the probability that the two letters are not the same.

- A. $\frac{1}{9}$
 B. $\frac{1}{3}$
 C. $\frac{4}{9}$

- D. $\frac{2}{3}$
 E. $\frac{8}{9}$

Label 32
 32. Four persons A, B, C, D sit randomly around a round table. The probability that A sits next to B is

- A. $\frac{1}{4}$
 B. $\frac{1}{3}$
 C. $\frac{1}{2}$
 D. $\frac{2}{3}$
 E. $\frac{5}{6}$

Label 33
 33. A die is thrown twice. Find the probability that the number obtained at the first throw is greater than that at the second throw.

- A. $\frac{1}{6}$
 B. $\frac{5}{12}$
 C. $\frac{1}{2}$
 D. $\frac{7}{12}$
 E. $\frac{5}{6}$

Label 34
 34. If $a : b = 3 : 2$, $b : c = 4 : 3$, then $a + b : b + c =$

- A. $7 : 10$
 B. $5 : 7$
 C. $1 : 1$
 D. $7 : 5$
 E. $10 : 7$

Label 35
35 Peter bought an article for \$ x . He sold it to Mary at a profit of 20%. Mary then sold it to John for \$90 at a loss of 25%. Find x .

- A. 56.25
B. 81
C. 90
D. 100
E. 144

Label 36
36 If x and y are integers with $x > y$, which of the following is/are true?

- I. $x^2 > y^2$
II. $\frac{1}{x} < \frac{1}{y}$
III. $10^x > 10^y$

- A. III only
B. I and II only
C. I and III only
D. II and III only
E. I, II and III

Label 37
37 Solve the inequality $x \log_{10} 0.1 > \log_{10} 10$

- A. $x > -1$
B. $x > 1$
C. $x > 100$
D. $x < 1$
E. $x < -1$

Label 38
38 If $x^2 + y^2 = 5$ and $x + y = 3$, then $x - y =$

- A. 1
B. -1
C. 1 or -1
D. 1 or -5
E. -1 or 5

Label 39
39

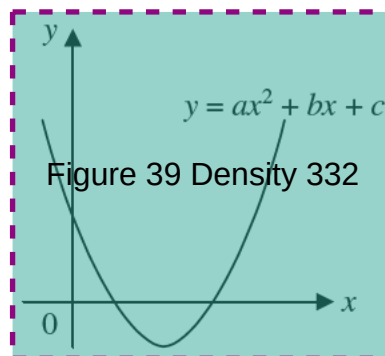


Figure 39 Density 332

The figure shows the graph of $y = ax^2 + bx + c$. Which of the following is/are true?

- I. $a > 0$
II. $b > 0$
III. $c > 0$

- A. I only
B. I and II only
C. I and III only
D. I and II only
E. I, II and III

Label 40
40 Find the H.C.F. of $(2x - 1)(x^2 - 6x + 9)$ and $(x^2 - 3x)(4x^2 - 1)$.

- A. $(x - 3)$
B. $(2x - 1)$
C. $(x - 3)(2x - 1)$
D. $x(x - 3)^2(2x - 1)(2x + 1)$
E. There is no H.C.F.

Label 41
41 If a is 10% less than b and b is 10% greater than c , then $a : c =$

- A. 1 : 1
B. 9 : 10
C. 10 : 9
D. 99 : 100
E. 100 : 99

Label 42
42 If $3a = 2b = 5c$, then $\frac{1}{a} : \frac{1}{b} : \frac{1}{c} =$

- A. 3 : 2 : 5
B. 5 : 2 : 3

- C. $\frac{1}{3} : \frac{1}{2} : \frac{1}{5}$
 D. $\frac{1}{5} : \frac{1}{3} : \frac{1}{2}$
 E. $\frac{1}{2} : \frac{1}{3} : \frac{1}{5}$

Label 43
 43. A man walks from place A to place B at a speed of 3 km/h and cycles immediately back to place A along the same road at a speed of 15 km/h. The average speed for the whole trip is

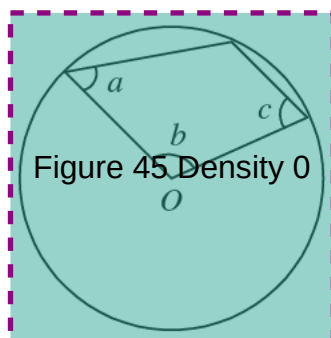
- A. 3 km/h.
 B. 6 km/h.
 C. 9 km/h.
 D. 10 km/h.
 E. 12 km/h.

Label 44
 44. Let n be a positive integer. Which of the following number is/are odd?

- I. 2^{2n+1}
 II. $2^n + 1$
 III. $3(2^n)$

- A. I only
 B. II only
 C. III only
 D. II and III only
 E. I, II and III

Label 45
 45.

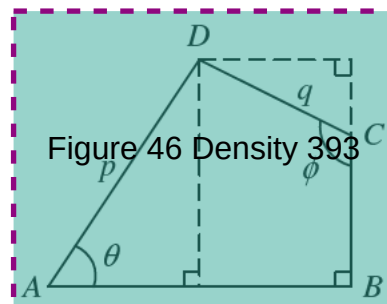


In the figure, O is the center of the circle. $a + b =$

- A. 180°
 B. c
 C. c
 D. $\frac{c}{2}$

- D. $180^\circ - c$
 E. $180^\circ - \frac{c}{2}$

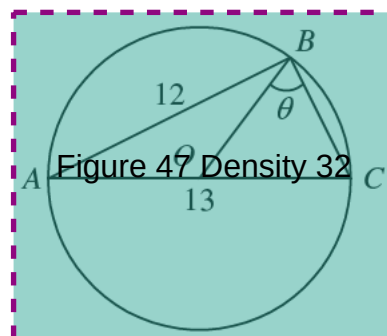
Label 46
 46.



In the figure, $AD = p$, $CD = q$ and $\angle B = 90^\circ$. $BC =$

- A. $p \sin \theta - q \sin \phi$
 B. $p \sin \theta - q \cos \phi$
 C. $p \cos \theta - q \sin \phi$
 D. $p \sin \theta + q \cos \phi$
 E. $p \cos \theta + q \sin \phi$

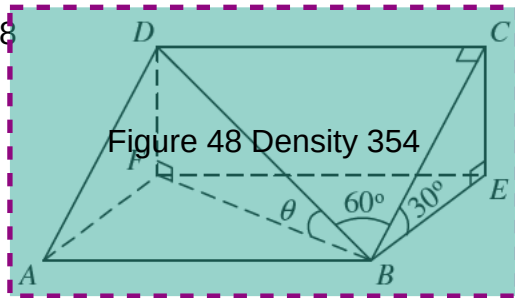
Label 47
 47.



In the figure, O is the centre of the circle. If $AB = 12$ and $AC = 13$, then $\cos \theta =$

- A. $\frac{5}{12}$
 B. $\frac{5}{13}$
 C. $\frac{12}{13}$
 D. $\frac{12}{25}$
 E. $\frac{13}{25}$

Label 48



In the figure, $ABCD$ is a rectangle inclined at an angle of 30° to the horizontal plane $ABEF$. $\angle CBD = 60^\circ$. Let θ be the inclination of BD to the horizontal plane. $\sin \theta =$

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. $\frac{\sqrt{3}}{2}$
- D. $\frac{\sqrt{3}}{3}$
- E. $\frac{\sqrt{3}}{4}$

Label 49 How many different values of x between 0° and 360° will satisfy the equation $(\sin x + 1)(2 \sin x + 1) = 0$?

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

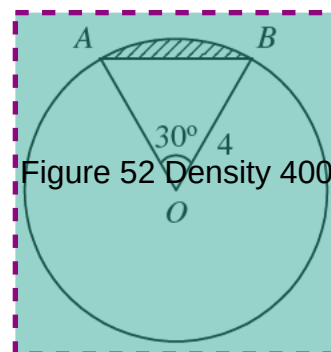
Label 50 If $0^\circ \leq x < 360^\circ$, the number of points of intersection of the graph of $y = \sin x$ and $y = 1 + \cos x$ is

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

Label 51 In $\triangle ABC$, if $AB : BC : CA = 4 : 5 : 6$, then $\cos A =$

- A. $\frac{1}{8}$
- B. $\frac{1}{5}$
- C. $\frac{3}{10}$
- D. $\frac{9}{16}$
- E. $\frac{3}{4}$

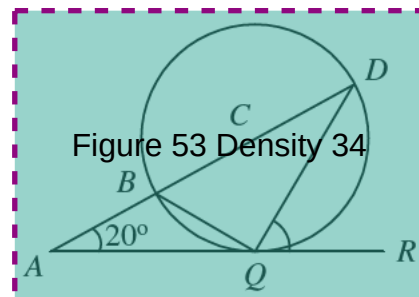
Label 52



In the figure, O is the centre of the circle of radius 4. The area of the shaded region is

- A. $\frac{4\pi}{3} - 4$
- B. $\frac{4\pi}{3} - 8$
- C. $\frac{4\pi}{3} - 4\sqrt{3}$
- D. $\frac{2\pi}{3} - 4$
- E. $\frac{8\pi}{3} - 8$

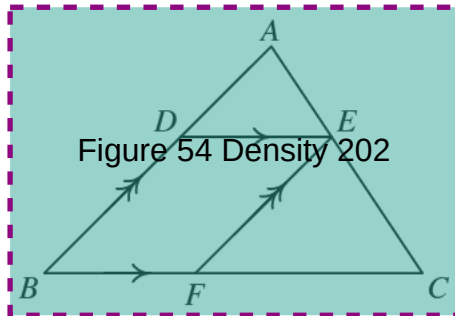
Label 53



In the figure, C is the centre of the circle. $ABCD$ is a straight line. AQR touches the circle at Q . If $\angle DAR = 20^\circ$, then $\angle DQR =$

- A. 35°
- B. 40°
- C. 55°
- D. 65°
- E. 70°

Label 54



In the figure, $DE \parallel BC$ and $AB \parallel EF$. If $AE : EC = 1 : 2$, then area of $\triangle ADE$: area of parallelogram $BFED =$

- A. $1 : 2$
- B. $1 : 3$
- C. $1 : 4$
- D. $1 : 5$
- E. $1 : 6$